Reflection and Transmission of Sound Waves

1. Introduction

Sound waves are a major topic in Physics. They are **mechanical longitudinal waves** that require a material medium (air, liquid, or solid) to propagate.

When a sound wave encounters a boundary between two different media, three main phenomena occur:

- Reflection
- Transmission
- Absorption

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2. Reflection of Sound Waves

2.1 Definition

Reflection is the bouncing back of a sound wave into the original medium when it strikes a surface or boundary.

2.2 Laws of Reflection

Sound reflection follows the same principle as light, described by Law of Reflection:

- The angle of incidence equals the angle of reflection
- The incident wave, reflected wave, and the normal lie in the same plane

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2.3 Types of Reflection

1. Specular Reflection

- Occurs on smooth, hard surfaces
- Produces clear and directed reflected waves

2. Diffuse Reflection

- Occurs on rough or irregular surfaces
- Scatters sound in multiple directions

2.4 Echo and Reverberation

- **Echo**: A distinct repetition of sound due to reflection
- **Reverberation**: Persistence of sound caused by multiple reflections

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2.5 Factors Affecting Reflection

- Nature of the surface (hard vs soft)
- Size of the obstacle
- Frequency of the sound wave

2.6 Applications

- SONAR (Sound Navigation and Ranging)
- Architectural acoustics (concert halls, theaters)
- Sound insulation design

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3. Transmission of Sound Waves

3.1 Definition

Transmission is the passage of a sound wave from one medium into another across a boundary.

3.2 What Happens During Transmission?

When sound crosses a boundary:

- Part of the wave is transmitted
- Speed may change
- Direction may change (related to Refraction)
- Intensity is reduced

3.3 Reflection and Transmission Coefficients

At a boundary:

- Reflection coefficient (R): fraction of reflected energy
- Transmission coefficient (T): fraction of transmitted energy

$R + T = 1$ (if no absorption)

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4. Energy Distribution at a Boundary

When a sound wave hits a boundary:

- Reflected energy
- Transmitted energy
- Absorbed energy

$$\text{Total Energy} = R + T + A$$

Where (A) is absorption.

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5. Comparison Between Reflection and Transmission

Property	Reflection	Transmission
Direction	Returns to original medium	Passes into new medium
Energy	Reflected portion	Transmitted portion
Depends on	Surface properties	Acoustic impedance
Result	Echo/reverberation	Sound propagation through materials

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Standing waves and resonance

Standing waves

A standing wave is a wave formed by the interference of two waves that have the same frequency and amplitude and travel in opposite directions. It appears to be stationary in space, with fixed points called nodes and points of maximum oscillation called antinodes.

Derivation of the Standing Wave Equation

A standing wave is formed by the superposition of two waves that:

- Have the same amplitude (A)
- Have the same frequency (f)
- Travel in opposite directions

Write the Two Traveling Waves

Wave moving to the right:

$$y_1 = A \sin(kx - \omega t)$$

Wave moving to the left:

$$y_2 = A \sin(kx + \omega t)$$

Where:

- (A) = amplitude
- (k = $\frac{2\pi}{\lambda}$ }{\lambda}) (wave number)
- (ω= 2π f) (angular frequency)

Apply the Principle of Superposition

The resultant wave is:

$$y = y_1 + y_2$$
$$y = A \sin(kx - \omega t) + A \sin(kx + \omega t)$$
$$y = A [\sin(kx - \omega t) + \sin(kx + \omega t)]$$

Use a Trigonometric Identity

We use the identity:

$\sin(a - b) + \sin(a + b) = 2 \sin(a)\cos(b)$

Let:

- (a = kx)
- (b = ω t)

Final Result

$y = 2A \sin(kx)\cos(\omega t)$
This is the **standing wave equation

Physical Meaning of the Equation

✳ **Spatial part:** sin(kx)

- Depends only on position (x)
- Determines **nodes and antinodes**

✳**Time part:** cos(ωt)
]**.** Depends only on time (t)

. Describes oscillation in time

Resonance

Definition of Resonance

Resonance is a physical phenomenon that occurs when a system is driven by an external periodic force whose frequency matches the system's natural frequency. This causes a large increase in the amplitude of oscillation.

condition:

$$f_{external} = f_{natural}$$

Relationship between Resonance and Standing Waves

Resonance is the physical cause, while standing waves are the resulting pattern.

How they are connected:

1. A wave is produced in a medium (string or air column)
2. It reflects from boundaries
3. The incident and reflected waves interfere
4. At specific frequencies (resonant frequencies), the interference becomes stable
5. A standing wave pattern is formed

Therefore:
Standing waves occur only at resonant frequencies

Comparison: Resonance vs Standing Waves

Feature	Resonance	Standing Waves
Definition	Large increase in amplitude at natural frequency	Fixed wave pattern formed by interference
Type	Physical phenomenon	Wave pattern
Cause	Matching external and natural frequency	Interference of two opposite waves
Result	Increased energy and amplitude	Nodes and antinodes formation
Relationship	Causes standing waves	Result of resonance in many systems

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Waves phenomena

1.Interference and Superposition Principle

Superposition Principle

Definition:

The superposition principle states that when two or more waves travel through the same medium and overlap at a point, the resultant displacement at that point is equal to the algebraic sum of the displacements produced by each individual wave.

$$y= y_1 + y_2 + y_3 + ..$$

Conditions for Superposition:

- Waves must be in the same medium
- Waves must overlap in space and time
- Medium must be linear (behaves normally under small disturbances

Interference

Definition:

Interference is the phenomenon that occurs when two or more waves overlap and combine due to the superposition principle, producing a new resultant wave pattern.

Explanation:

- Interference is not a separate law; it is a direct result of superposition.
- The resulting wave depends on how the individual waves align.

Types of Interference

1) Constructive Interference

Occurs when waves meet **in phase** (crest meets crest or trough meets trough).

- Amplitudes add up

$$A = A_1 + A_2$$

Produces a larger wave
Increases intensity (sound becomes stronger)

2) Destructive Interference

Occurs when waves meet out of phase (crest meets trough).

- Amplitudes subtract

  $A = A_1 - A_2$

Produces a smaller wave
Can cancel completely if amplitudes are equal

Relationship Between Superposition and Interference

- **Superposition principle** → explains how waves combine mathematically
- **Interference** → describes the physical result of that combination

Therefore:
Interference is the observable effect of superposition.

Real-Life Examples

Sound waves:

- Noise cancellation headphones use destructive interference.

Light waves:

- Colors on soap bubbles are due to interference.

2.Diffraction

Definition

Diffraction is the phenomenon in which waves bend and spread out when they pass through a narrow opening or around an obstacle.

Example

A clear example of diffraction is:

- Hearing someone talking clearly even when they are behind a wall or around a corner.

Conditions for Diffraction

Diffraction becomes noticeable when:

Size of opening or obstacle$\approx \lambda$

Where:

- (λ) = wavelength

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conditions:

1. The wavelength should be comparable to the size of the gap or obstacle
2. Smaller openings produce stronger diffraction
3. Larger wavelengths show more noticeable diffraction

3.Refraction and Snell's Law

Refraction

Definition:

Refraction is the change in direction of a wave when it passes from one medium to another with a different optical density, due to a change in its speed.

When a wave enters a new medium:

- Its speed changes
- So it bends (changes direction)

Examples:

- A pencil appearing bent in water
- Light bending when it passes from air into water or glass

Snell's Law

Definition:

Snell's Law describes the relationship between the angles and speeds of a wave when it passes between two different media.

$$\frac{\sin\theta\,1}{\sin\theta 2} = \frac{n1}{n2}$$

(n_1, n_2): refractive indices of the two media

- $\theta 1$: angle of incidence
- $\theta 2$: angle of refraction

What happens during refraction?

- When entering a **denser medium**:
  The wave slows down
  It bends toward the normal
- When entering a **less dense medium**:
  The wave speeds up
  It bends away from the normal

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4. <u>Attenuation and Absorption</u>

1.Attenuation

Definition:

Attenuation is the gradual decrease in the intensity (or amplitude) of a wave as it travels through a medium.

As a wave moves forward:

- it becomes weaker
- its energy spreads out or is lost

Causes of attenuation:

- Spreading of energy over distance
- Scattering by particles
- Conversion of wave energy into heat

Example:

- Sound becoming quieter as you move away from the source
- Weakening of seismic waves inside Earth

2. Absorption

Definition:

Absorption is the process in which a medium takes in the energy of a wave and converts it into other forms of energy, usually heat.

Instead of passing through or reflecting:

- the wave energy is “taken in” by the material

Examples:

- Dark clothes absorbing sunlight and heating up
- Soundproof materials absorbing sound waves
- Foam absorbing sound in recording studios

Relationship between them

- **Absorption** is one of the main causes of **attenuation**
- Attenuation is the overall reduction in wave strength
- Absorption is the process of energy loss inside the medium